

Data Structures and Algorithms · 21CSC201J

Unit 1 · Session 2 · SLO 2 — Solved Worksheet

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Unit 1 : Session 2 : SLO 2 | Array of Structures

I. Fill in the Blanks

- multiple records → an array of structures stores many records of the same type.
- arrayName[size] → struct StructName arr[100];
- arr[index].member → access a member via index and dot operator.
- 5 Student structures → struct Student s[5]; declares an array of 5 Student records.
- student/employee records → e.g. roll numbers, names, salaries.
- loop (for loop) → read values into the array with a loop and scanf.
- emp[2].name → the 3rd element's name field (index 2).
- normal structure variable → each element behaves like a regular struct variable.
- arr (or &arr[0]) → struct StructName *ptr = arr;
- for loop → loop with the dot operator prints each record's details.

II. Code Snippets

- 100 students: struct Student { int roll; char name[50]; }; struct Student s[100]; → loop i=0..99 with scanf("%d %s", &s[i].roll, s[i].name).
- 5 employees: struct Emp { int id; float salary; }; struct Emp e[5]; → input loop then display loop printing e[i].id, e[i].salary.
- 10 students → 3 subjects: struct Result { int marks[3]; }; struct Result r[10]; → r[i].marks[j] holds subject j of student i.
- Shopping products: struct Product { char name[50]; float price; }; struct Product p[50];
- Cricket team: struct Player { char name[50]; int runs; }; struct Player team[11];

III. Practice the Code

- struct Car { char model[30]; int year; };
- struct Car c[3] = {{"Swift", 2020}, {"City", 2022}, {"Creta", 2023}};
- for (int i = 0; i < 3; i++) printf("%s (%d)\n", c[i].model, c[i].year);